

About the Prediction Formation Standard

What Is the Prediction Formation Standard?

The Prediction Formation Standard (PFOS) is the Parent Standard within PGA™ — Prediction Governance Architecture that governs how the components required for prediction become a specific Governed Prediction .

Before Prediction Formation, a predictive process may already possess:

a defined Prediction Object and Purpose; a documented Prediction Methodology;

a governed Prediction Basis; governed Predictive Assumptions; applicable predictive conditions; model, analytical, expert, or other predictive outputs.

These components do not automatically constitute a prediction.

PFOS establishes the governed formation process through which they become one identifiable, bounded, traceable, and historically reconstructable predictive claim.

Its canonical transition is:

Governed Prediction Object & Purpose + Governed Prediction Methodology + Governed Prediction Basis + Governed Predictive Assumptions + Applicable Conditions → Governed Prediction

PFOS governs the formation of the prediction.

It does not determine whether the prediction will become true.

Why the Standard Exists

Organizations increasingly generate predictive outputs through:

AI systems; statistical models; forecasting engines; expert analysis; financial models; simulations; prediction markets; human-AI workflows; automated decision systems.

Yet producing an output and forming a governed prediction are not the same thing.

A system may produce:

72%

10,000 units

failure expected

market likely to decline

without establishing precisely:

what is being predicted; what the predictive claim actually means; over what Prediction Horizon it applies; which Predictive Boundary applies; which methodology produced it; which information supported it; which assumptions were active; which conditions applied; when the prediction was formed.

Without these relationships, the output may be difficult or impossible to legitimately interpret, monitor, revise, compare with an outcome, or assess for predictive performance.

PFOS exists to prevent this structural gap.



Prediction Output ≠ Governed Prediction

What PFOS Governs

PFOS governs the formation of a specific prediction from governed predictive components.

This includes governance of:

Prediction Composition; Predictive Claim identification; Prediction Output Formation; Prediction State establishment; Formation Time; Prediction Identity; Prediction Object association; Prediction Purpose association; Prediction Horizon association; Predictive Boundary association; Prediction Methodology association; Prediction Basis association; Predictive Assumption association; applicable Formation Conditions; conditionality; formation provenance; human, machine, and hybrid formation; Formation Logic; Prediction Formation Gaps; formation restrictions; formation traceability; Formation-Time historical integrity.

The result is a prediction that can remain identifiable throughout the rest of its lifecycle.

Predictive Components ≠ Governed Prediction

A documented methodology does not itself constitute a prediction.

A dataset does not constitute a prediction.

An assumption set does not constitute a prediction.

A model output does not necessarily constitute a prediction.

Even when all of these elements exist simultaneously, the predictive system still needs to establish the specific claim created from them.

PFOS governs that transition.

Methodology + Basis + Assumptions + Output ≠ Governed Prediction by Default

Prediction Output ≠ Predictive Claim

A predictive system may produce an output such as:

0.82

That output alone may not establish whether it represents:

82% probability of an event; 82% confidence; an 82% risk score;

normalized model output; threshold likelihood; classification score; another computational result.

PFOS requires the specific Predictive Claim to be identifiable.

For example:

The governed prediction states that Event X is predicted to occur within Horizon Y under Conditions A, B, and C.

The meaning of the prediction must not have to be reconstructed from the number after the fact.

What Is a Governed Prediction?

A Governed Prediction is a specific predictive claim formed through sufficiently governed predictive components and retained with enough structure to establish:

what was predicted

for what purpose

over what horizon

within what boundary

using what methodology
from what Prediction Basis
under what assumptions
subject to what conditions
and:

when the prediction was formed.

A Governed Prediction may ultimately be correct or incorrect.

Governance does not guarantee predictive success.

Governed Prediction $\neq$ Correct Prediction

The Prediction Formation Event

PFOS establishes a distinguishable Prediction Formation Event .

This represents the point at which predictive preparation becomes a specific prediction.

Before formation, the system may contain:

candidate predictions + model outputs + scenarios + analytical results

After formation, it contains:

an identifiable Governed Prediction.

This distinction becomes essential later when determining:

which prediction was actually active; what information existed when it was formed; which assumptions applied;

whether it was later revised; which prediction should be compared with the eventual outcome.

Formation Time Matters

A prediction must remain associated with its Formation Time.

Formation Time is not necessarily:

publication time; communication time; reliance time; outcome time.

A prediction may be formed internally on Monday and published on Wednesday.

The predictive conditions available on Monday belong to the formation record.

Formation Time $\neq$ Publication Time

This prevents later information from being silently incorporated into an earlier prediction.

Conditional Predictions Must Remain Conditional

Many legitimate predictions depend upon explicit conditions.

For example:

If supplier capacity remains above 80%, production is predicted to exceed 50,000 units during Q4.

Removing the condition changes the prediction:

Production will exceed 50,000 units during Q4.

These statements are not equivalent.

PFOS therefore preserves conditionality as part of Prediction Formation.

Conditional Prediction $\neq$ Unconditional Prediction

Human, Machine, and Hybrid Prediction Formation

PFOS is technology-neutral.

A prediction may originate through:

Human Formation

An analyst or expert constructs the prediction.

Machine Formation

An AI model, statistical model, forecasting engine, or automated system constructs the predictive output.

Hybrid Formation

Machine output is combined with human judgment, adjustment, selection, or additional conditions.

Governance does not depend upon who or what created the prediction.

The relevant question is whether the final predictive claim remains sufficiently governed.

Machine-Generated Prediction $\neq$ Governed Prediction by Default

Candidate Predictions and the Final Prediction

Predictive processes may generate several possible outputs.

For example:

Model A $\rightarrow$ 4.2% growth

Model B $\rightarrow$ 6.1% growth

Expert Adjustment $\rightarrow$ 5.4% growth

Stress Scenario $\rightarrow$ 1.8% growth

PFOS requires these outputs to remain distinguishable where material.

If one becomes the official prediction, the formation process must establish which one.

Multiple Candidate Predictions $\neq$ One Governed Prediction

This becomes especially important when predictive performance is later assessed.

Prediction Formation State

PFOS allows the formation process to preserve meaningful governance states.

Depending upon implementation, these may include:

Draft

Formed

Conditionally Formed

Formation Restricted

Formation Incomplete

Unable to Form

This prevents an incomplete predictive process from being represented as a fully governed prediction merely because an output exists.

Inability to Form Is a Legitimate Governance Result

Prediction governance should not force a prediction into existence.

A Governed Prediction may be unable to form where, for example:

Prediction Object remains ambiguous; methodology is insufficiently documented;

Prediction Basis is materially inadequate; a Critical Assumption remains hidden or unresolved; Prediction Horizon is undefined; predictive conditions materially conflict.

PFOS can therefore produce:

Unable to Form

without making any claim about the future itself.

Inability to Form a Governed Prediction $\neq$ Prediction of Failure

Formation-Time Integrity

Once a prediction is formed, its original predictive context must remain reconstructable.

Later changes to:

methodology; information; assumptions; conditions; probability; feasibility; prediction state

must not silently rewrite the original formation record.

A prediction evaluated six months later must still be reconstructable as it existed when formed.

Current Predictive State $\neq$ Formation-Time Predictive State

Position Within PGA™

PFOS occupies the fifth governed space within the Prediction Lifecycle:

Prediction Object & Purpose → Prediction Methodology → Prediction Basis → Prediction Assumptions → Prediction Formation → Probability & Uncertainty → Prediction Feasibility → Collective Prediction where applicable → Communication & Reliance → Monitoring & Revision → Outcome & Performance → Prediction Record & Continuity

Current Governed Space: Prediction Formation

PFOS represents a major transition within PGA™.

The preceding standards establish the predictive foundation.

PFOS turns that foundation into the specific predictive object that the remaining architecture governs.

How PFOS Is Operationalized

PFOS is operationalized through five Core Modules:

PCOM — Prediction Composition Module

Governs how the governed predictive components are assembled into a coherent prediction structure.

PCIM — Predictive Claim Identification Module

Governs identification and precise representation of the specific predictive claim.

POFM — Prediction Output Formation Module

Governs transformation of applicable predictive outputs and formation logic into the specific prediction output state.

PSEM — Prediction State Establishment Module

Governs establishment of the prediction's formation state, including formed, conditional, restricted, incomplete, or unable- to-form states.

PFTM — Prediction Formation Traceability Module

Governs traceability between the formed prediction and the methodology, Prediction Basis, assumptions, conditions, provenance, and Formation-Time state from which it originated.

The internal progression is:

Prediction Composition → Predictive Claim Identification → Prediction Output Formation → Prediction State Establishment → Prediction Formation Traceability

Relationship to the Prediction Methodology Standard

PMES — Prediction Methodology Standard governs the methodology by which predictions are constructed.

PFOS governs formation of the specific prediction using that methodology.

A methodology may exist without any prediction currently being formed.

Prediction Methodology ≠ Formed Prediction

Relationship to the Prediction Basis Standard

PBAS — Prediction Basis Standard governs what information, observations, signals, and governed references legitimately form the basis of the prediction.

PFOS associates the applicable Prediction Basis with the prediction at Formation Time.

Prediction Basis ≠ Predictive Claim

Relationship to the Prediction Assumption Standard

PAS — Prediction Assumption Standard governs the assumptions that must hold for the prediction to remain meaningful.

PFOS associates the applicable governed assumption state with the formed prediction.

Predictive Assumptions ≠ Prediction

Relationship to Probability & Uncertainty Governance

PFOS does not determine how probability, confidence, uncertainty, or predictive ranges should be represented.

That belongs to:

PPUS — Prediction Probability & Uncertainty Standard

PFOS forms the prediction that PPUS receives.

Prediction Formation ≠ Probability Representation

Relationship to Prediction Feasibility

A prediction may be coherently formed while the predicted outcome is later assessed as difficult or impossible to achieve.

Feasibility belongs to:

PFES — Prediction Feasibility Standard

Therefore:

Prediction Formation ≠ Prediction Feasibility

Relationship to Collective Prediction

PFOS can establish an individual Governed Prediction.

Where multiple predictions originating from participants, models, agents, markets, or forecasting mechanisms are combined, the legitimacy of that collective composition belongs to:

CPGS — Collective Prediction Governance Standard

Single Prediction Formation $\neq$ Collective Prediction Governance

Relationship to Monitoring & Revision

PFOS establishes the original prediction state.

Later review, revision, supersession, or withdrawal belongs to:

PMRS — Prediction Monitoring & Revision Standard

The original formation state must remain preserved even after revision.

Formation $\neq$ Revision

What PFOS Does Not Govern

PFOS does not govern:

definition of Prediction Object and Purpose;

Prediction Methodology design; Prediction Basis governance; Assumption Governance; evidence validation; Operational Reality determination; probability representation; uncertainty representation; final Prediction Feasibility; Collective Prediction legitimacy; prediction communication; reliance decisions; prediction-level monitoring; prediction revision; prediction supersession; prediction withdrawal; outcome determination; predictive performance assessment; complete lifecycle record continuity.

These spaces remain governed by their respective PGA™ standards or compatible OOF® architectures.

Cross-Architecture Boundaries

ORA™ — Operational Reality Architecture governs Operational Reality. PFOS may reference operational conditions but does not establish reality.

ADIT® governs evidence. PFOS may consume governed evidence references but does not validate evidence.

SIMULOS® governs simulation. Simulation outputs may support Prediction Formation, but:

Simulation Output $\neq$ Governed Prediction

GOA™ and AGA™ may govern organizational authority and accountability surrounding who or what is permitted to finalize predictions. PFOS preserves relevant formation references without absorbing those governance spaces.

AIG® and ASGA™ retain governance of AI and autonomous-system behavior where such systems participate in Prediction Formation.

Why PFOS Matters

Prediction governance becomes weak when organizations govern models and data but fail to govern the actual prediction produced from them.

Without PFOS, an organization may later know:

which model existed; which dataset existed; which assumptions existed;

while still being unable to establish:

what exactly was predicted.

PFOS creates the stable predictive object required for:

probability governance; feasibility assessment; communication; reliance; monitoring; revision;
outcome comparison; performance assessment; historical reconstruction.

It creates the point at which predictive analysis becomes a governable prediction.

Use Case 1 — Nobody Can Determine Which Forecast Was the Actual Prediction

A company prepares its annual revenue forecast using:

an AI forecasting model; a statistical model; regional analyst estimates; management adjustments; several scenario outputs.

The final executive presentation states:

Expected revenue growth: 6–10%.

Six months later, actual growth is 4%.

Management wants to evaluate forecasting performance.

One team claims the prediction was:

8.7%.

Another identifies:

7.1%.

A third argues that any value between 6% and 10% should count as correct.

No governed formation event established which output became the actual prediction.

The critical governance question is:

How can predictive performance be legitimately assessed when the organization cannot reconstruct what its prediction actually was?

PFOS establishes the specific Predictive Claim, Formation Time, applicable methodology, Prediction Basis, assumptions, conditions, and formation state.

Candidate outputs remain distinguishable from the final Governed Prediction.

The organization can later compare the actual outcome against the prediction that genuinely existed at Formation Time.

Multiple analytical outputs do not become one prediction merely because they appear in the same presentation.

Use Case 2 — An AI Dashboard Displays 82%, but Users Are Predicting the Wrong Event

An AI system displays:

82%

on an operational risk dashboard.

Managers interpret the number as:

82% probability that the project will fail within 90 days.

Operational decisions begin relying on that interpretation.

The technical team later explains that the model output actually represents:

an 82% modeled likelihood that a defined risk indicator will exceed a threshold within 90 days under specified system conditions.

The model never directly predicted project failure.

The critical governance question is:

Can an AI output legitimately function as an organizational prediction when the predictive claim behind that output has never been formally established?

PFOS requires the organization to distinguish the raw model output from the Predictive Claim and establish:

Prediction Object; Prediction Purpose; Prediction Horizon; Predictive Boundary; methodology; Prediction Basis; assumptions; conditions; Formation Time; formation state.

Only then can the prediction proceed into probability, feasibility, communication, and reliance governance.

A number is not a governed prediction until the claim behind the number is governable.

Canonical Boundary

PFOS governs the composition, identification, formation, state establishment, conditionality, provenance, timing, traceability, and historical Formation-Time integrity of a specific Governed Prediction created from governed Prediction Object and Purpose, Prediction Methodology, Prediction Basis, Predictive Assumptions, and applicable predictive conditions. It does not govern Prediction Methodology design, Prediction Basis governance, Assumption Governance, probability and uncertainty representation, final feasibility determination, Collective Prediction composition, communication and reliance, prediction-level monitoring and revision, outcome determination, predictive performance assessment, or lifecycle record continuity beyond the formation state.

PFOS governs:

how predictive components become a specific prediction.

It does not govern:

whether that prediction will become true.

Canonical Closing Statement

A prediction cannot be governed if nobody can establish what was actually predicted.

PFOS creates the governed transition from predictive components to one identifiable, bounded, traceable, and historically reconstructable predictive claim.

A governed prediction may be wrong. It must not be undefined at the moment it is formed.

Governed Links

[About Prediction Formation Standard \(PFOS\).](#)

[Prediction Component Organization Module \(PCOM\).](#)

[Prediction Condition Integration Module \(PCIM\).](#)

[Prediction Output Formation Module \(POFM\).](#)

[Prediction State Establishment Module \(PSEM\).](#)

[Prediction Formation Traceability Module \(PFTM\).](#)

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